

Day 36:

(IV) match case statement :-

- => Here "match case" is one of the new feature in python 3.10 version onwards.
- => match case statement is recommended to take in deciding pre-designed conditions in menu driven Applications.

Syntax:

```
match (choice Expr) :
    case choice label 1 :
        Block of statements-1
    case choice label 2 :
        Block of statements-2
    case choice label 3 :
        Block of statements-3
    ...
    case choice label-n :
        Block of statements-n
    case _ :
        default block of statements
    other statements in program
```

- => Here "match" and "case" are the keywords.
- => "choice Expr" represents either int or str or bool.
- => if "choice Expr" is matching with "case label 1" then pvm executes block of statements-1 and later executes other statements in program.
- => In General "choice Expr" is trying match with case label-1, case label-2, case label-n then pvm executes corresponding block of statements and later executes other statements in program.
- => if "choice Expr" is not matching with any of the specified case labels then pvm executes default block of statement which are written under default case block (case _) and later executes other statements in program.

=> Writing default case block is optional and if we write then it must be written at last (otherwise we get syntax error).

=> When we represent multiple case labels under one case then those case labels must be combined with Bitwise OR operator (|)

program implementing all Arithmetic operations by using match case.

```
print("=" * 50)
print("\t Arithmetic operations")
print("=" * 50)
print("\t1. Addition")
print("\t2. Subtraction")
print("\t3. Multiplication")
print("\t4. Division")
print("\t5. Floor division")
print("\t6. modulo division")
print("\t7. exponentiation")
print("\t8. exit")
print("=" * 50)
ch = int(input("Enter your choice :"))
match(ch):
    case 1:
        print("Enter two values for Addition:")
        a, b = int(input()), int(input())
        print("sum ({} , {}) = {}".format(a, b, a+b))
    case 2:
        print("Enter two values for subtraction:")
        a, b = int(input()), int(input())
        print("sub ({} , {}) = {}".format(a, b, a-b))
```

Case 3:

```
print("Enter two values for mul:")
k, v = int(input()), int(input())
print("mul(12, 22) = 22".format(k, v, k*v))
```

Case 4:

```
print("Enter two values for Div:")
k, v = int(input()), int(input())
print("Div(12, 22) = 22".format(k, v, k/v))
```

Case 5:

```
print("Enter two value for floor Div:")
k, v = int(input()), int(input())
print("floor Div(12, 22) = 22".format(k, v, k//v))
```

Case 6:

```
print("Enter two values for mod Div:")
k, v = int(input()), int(input())
print("mod(12, 22) = 22".format(k, v, k%v))
```

Case 7:

```
k, x = int(input("Enter Base:")), int(input("Enter power"))
print("pow(12, 22) = 22".format(k, v, k**v))
```

Case 8:

```
print("Thx for using program")
```

```
case_ : # default case block
```

```
print("ur selection of operation is wrong - try again")
```

```
print("program execution completed")
```

Program :-

```

wkn = input("Enter week name")
match (wkn.upper()):
    case "MONDAY":
        print("{} is working day".format(wkn))
    case "TUESDAY":
        print("{} is working day".format(wkn))
    case "WEDNESDAY":
        print
    case "THURSDAY":
        print
    case "FRIDAY":
        print
    case "SATURDAY":
        print
    case "SUNDAY":
        print
    case _:
        print("{} is not a week day".format(wkn))

```

(Capital letter)

Program 3:-

```

wkn = input("Enter week name:")
match (wkn.upper()):
    case "MONDAY" | "TUESDAY" | "WEDNESDAY" | "THURSDAY" | "FRIDAY":
        print("{} is working day".format(wkn))
    case "SATURDAY":
        print("{} is weekend".format(wkn))
    case "SUNDAY":
        print("{} is Holiday".format(wkn))
    case _:
        print("{} is not a week day".format(wkn))

```



```

problem ④
wkn=input("Enter week name:")
if wkn.upper() in ["MON","TUE","WED","THU","FRI","SAT",
                  "SUN","MONDAY","TUESDAY","WEDNESDAY",
                  "THURSDAY","FRIDAY","SATURDAY",
                  "SUNDAY"]:
    match(wkn[0:3].upper()):
        case "MON"|"TUE"|"WED"|"THU"|"FRI":
            print("{ } is working day".format(wkn))
        case "SAT":
            print("{ } is week end".format(wkn))
        case "SUN":
            print("{ } is Holiday".format(wkn))
else:
    print("{ } is not a week day".format(wkn))

```

program implementing all arithmetic operations by using match case

```

In [5]: print("="*50)
print("\t Arithmetic operations")
print("="*50)
print("\t1.Addition")
print("\t2.Substraction")
print("\t3.Multiplication")
print("\t4.Division")
print("\t5.Floor Division")
print("\t6.modulo Division")
print("\t7.Exponentiation")
print("\t8.Exit")
ch=int(input("enter your choice:"))
match(ch):
    case 1:
        print("Enter two values for addition")
        a,b=int(input()),int(input())
        print("sum({},{})={}".format(a,b,a+b))
    case 2:
        print("Enter two values for subtraction")
        a,b=int(input()),int(input())
        print("sub({},{})={}".format(a,b,a-b))
    case 3:
        print("Enter two values for multiplication")
        a,b=int(input()),int(input())
        print("mul({},{})={}".format(a,b,a*b))
    case 4:
        print("Enter two values for Division")
        a,b=int(input()),int(input())
        print("Div({},{})={}".format(a,b,a/b))
    case 5:
        print("Enter two values for Floor Division")
        a,b=int(input()),int(input())
        print("floor Div({},{})={}".format(a,b,a//b))

```

```

case 6:
    print("Enter two values for Modulo Division")
    a,b=int(input()),int(input())
    print("modulo Div({},{})={}".format(a,b,a%b))
case 7:
    print("Enter two Values for Exponentiation")
    a,b=int(input()),int(input())
    print("power({},{})={}".format(a,b,a**b))

```

```

=====
        Arithmetic operations
=====
1.Addition
2.Substraction
3.Multiplication
4.Division
5.Floor Division
6.modulo Division
7.Exponentiation
8.Exit
Enter two Values for Exponentiation
power(7,7)=823543

```

```

In [5]: wkn=input("Enter weak name:")
match(wkn.upper()):
    case "MONDAY":
        print("{} is a working day".format(wkn))
    case "TUESDAY":
        print("{} is a working day".format(wkn))
    case "WEDNESDAY":
        print("{} is a working day".format(wkn))
    case "THURSDAY":
        print("{} is a working day".format(wkn))
    case "FRIDAY":
        print("{} is a working day".format(wkn))
    case "SATURDAY":
        print("{} is a working day".format(wkn))
    case "SUNDAY":
        print("{} is off day".format(wkn))
    case _:
        print("{} is not a week day".format(wkn))

```

MONDAY is a working day

```

In [9]: wkn=input("enter week name:")
match(wkn.upper()):
    case "MONDAY" | "TUESDAY" | "WEDNESDAY" | "THURSDAY" | "FRIDAY":
        print({},is working day".format(wkn))
    case "SATURDAY":
        print({},is weekend".format(wkn))
    case "SUNDAY":
        print("{} ,is holiday".format(wkn))
    case _:
        print("{} is a not week day".format(wkn))

```

SUNDAY ,is holiday

```

In [10]: wkn=input("enter week name:")
if wkn.upper() in ["MON", "TUE", "WED", "THU", "FRI", "SAT", "SUN", "MONDAY", "TUESDAY",
    match(wkn[0:3].upper()):
        case "MON" | "TUE" | "WED" | "THU" | "FRI":

```

```
        print("{} is working day".format(wkn))
    case "SAT":
        print("{} is week end".format(wkn))
    case "SUN":
        print("{} is holiday".format(wkn))
else:
    print("{} is not a week day".format(wkn))
```

MON is working day

In []: